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Maks

$Z = 15000x_1 + 10000x_2$

s/t

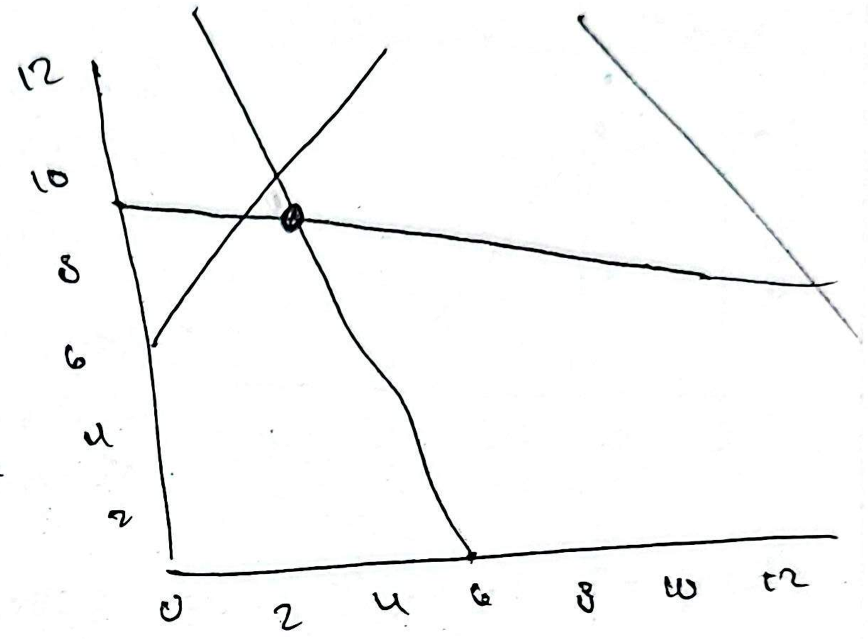
$6x_1 + 2x_2 \le 36$

$2x_1 + 15/2\, x_2 \le 72$

$3x_1 + 2x_2 \le 80$

$-x_1 + x_2 \le 6$

$x_1, x_2 \ge 0$ integer



Grafis

① $6x_1 + 2x_2 \le 36$

$x_1 = 0,\ x_2 = 18$ (0, 18)

$x_1 = 6,\ x_2 = 0$ ~~[illegible]~~ (6, 0)

② $2x_1 + 15/2\, x_2 \le$ ~~[illegible]~~ 72

$x_1 = 0,\ x_2 =$ ~~[illegible]~~ ~~[illegible]~~ 48/5 (0, 48/5) $(0, 9\frac{3}{5})$

$x_1 = 36,\ x_2 = 0$ ~~[illegible]~~ (36, 0)

③ $3x_1 + 2x_2 \le 80$

$x_1 = 0,\ x_2 = 40$ (0, 40)

$x_1 = 80/3,\ x_2 = 0$ (80/3, 0) $(26\frac{2}{3}, 0)$

④ ~~[illegible]~~ $-x_1 + x_2 \le 6$

$x_1 = 0,\ x_2 = 6$ (0, 6)

$x_1 = -6,\ x_2 = 0$ (−6, 0)

Z maks =

$15000x_1 + 10000x_2$

$= 133{,}902$

$\boxed{x_2 = 8\frac{32}{41}}$

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① $2x_2 = 36 - 6x_1 \rightarrow x_2 = 18 - 3x_1$

$\hookrightarrow 4x_1 + 15(18 - 3x_1) = 144$

$4x_1 + 270 - 45x_1 = 144$

$-41x_1 = -126$

$x_1 = 126/41$

$\boxed{x_1 = 3\frac{3}{41}}$

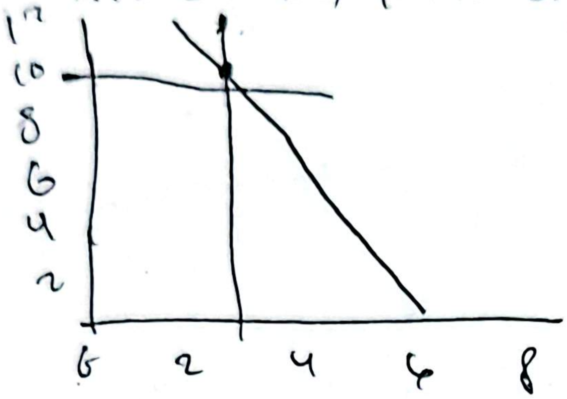
$x_2 = 18 - 3\left(3\frac{3}{41}\right) = \frac{738 - 378}{41} = \frac{360}{41}$

Dipindai dengan CamScanner

PL2

$x_1 = 3{,}073 \rightarrow$ cabang $x_1 \le 3$ dan $x_1 \ge 4$

$x_1 \le 3 \longrightarrow$ kendala baru



$\hookrightarrow 6x_1 + 2x_2 = 36$

$\hookrightarrow 6(3) + 2x_2 = 36$

$2x_2 = 18 \rightarrow \boxed{x_2 = 9}$

$4(3) + 15(9) =$

$12 + 135 = 147$

$\downarrow$

> 144

melebihi anggaran

$\hookrightarrow 4x_1 + 15x_2 = 144$

$4(3) + 15x_2 = 144$

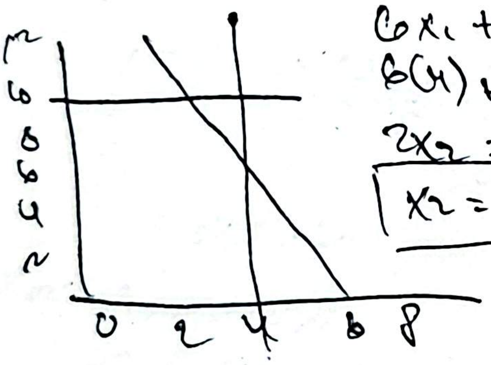
$15x_2 = 132 \rightarrow \boxed{x_2 = 8{,}8}$

$\hookrightarrow x_1 = 3,\ x_2 = 8{,}8 \longrightarrow z = 15\,000(3) + 10000(8{,}8)$

$z = 133{,}000$

PL3

$x_1 \ge 4 \longrightarrow$ kendala baru



$6x_1 + 2x_2 = 36$

$6(4) + 2x_2 = 36$

$2x_2 = 12$

$\boxed{x_2 = 6}$

$4x_1 + 15x_2 = 144$

$4(4) + 15(6) = 16 + 90 = 106$

masih ≤ 144 aman

$\hookrightarrow x_1 = 4,\ x_2 = 6$

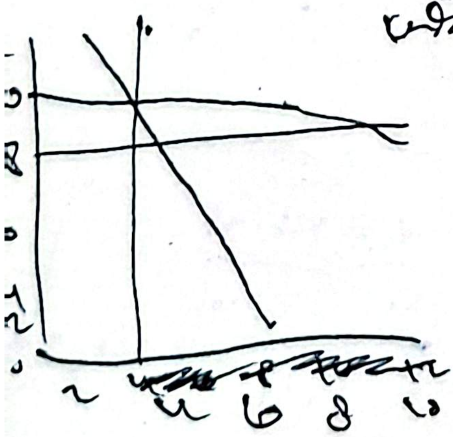
$z = 15\,000(4) + 10000(6) = 120.000$

$\therefore$ Batas awal lower bound

PL4

$x_2 = 8{,}8 \rightarrow$ cabang $\boxed{x_2 \le 8}$ dan $x_2 \ge 9$

kendala baru



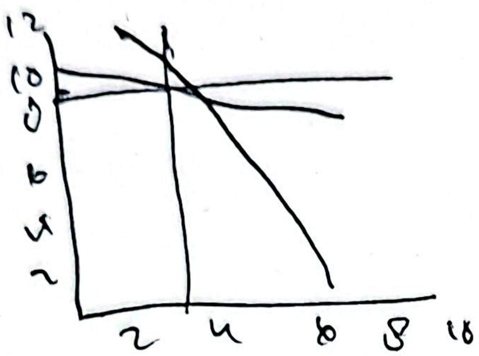
$6x_1 + 2(8) = 36$

$6x_1 = 20 \rightarrow x_1 = 3{,}33$ tidak sesuai dengan $x_1 \le 3$ jadi tetap $x_1 = 3$

$x_1 = 3,\ x_2 = 8$

$z = 15000(3) + 10000(8) = 125.000$

PL5 $x_1 \leq 3$, $x_2 \geq 9$



$6x_1 + 2(9) = 36$

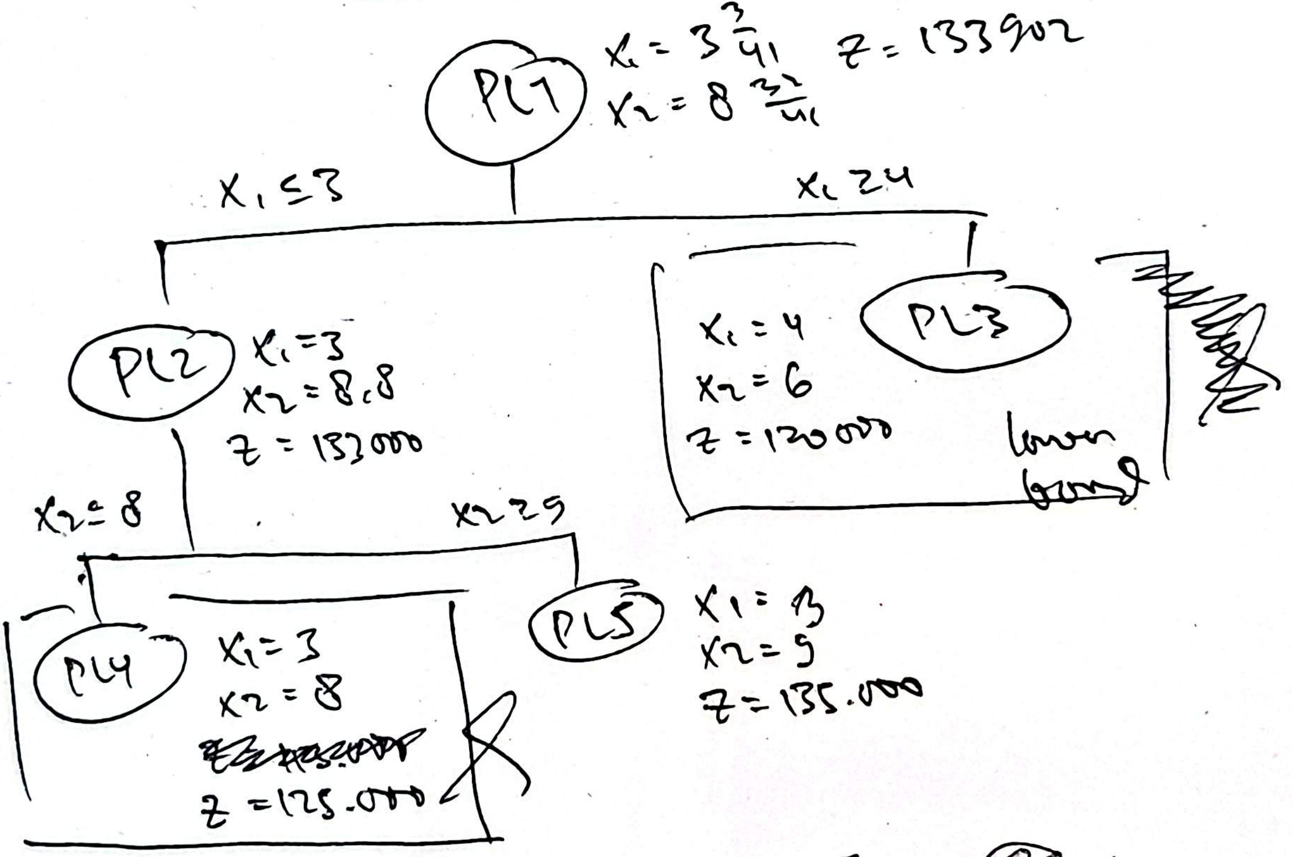
$6x_1 = 18 \rightarrow x_1 = 3$

$x_1 = 3, \ x_2 = 9$

$Z = 15000(3) + 10000(9)$

$= 135000$

cek constraint 2 : $4(3) + 15(9) = 147 > 144$
maka tidak layak


PL1
$x_1 = 3\frac{3}{41}$ $Z = 133902$
$x_2 = 8\frac{32}{41}$
$x_1 \leq 3$
$x_1 \geq 4$
PL2
$x_1 = 3$
$x_2 = 8,8$
$Z = 133000$
PL3
$x_1 = 4$
$x_2 = 6$
$Z = 120000$
lower bound
$x_2 \leq 8$
$x_2 \geq 9$
PL4
$x_1 = 3$
$x_2 = 8$
$Z = 125.000$
PL5
$x_1 = 3$
$x_2 = 9$
$Z = 135.000$

∴ Solusi optimal → $x_1 = 3$, $x_2 = 8$

Z maks = ~~[illegible]~~ 125.000

~~PL3~~ PL4

PL1 = hasil belum integer, belum fathom

PL2 : var keputusan belum integer, belum fathom

PL3 : sudah fathom namun lower bound

PL5 : tidak masuk di constraint 2, tidak fathom